Signals and Systems
SET 214
Lecture (10)- Laplace

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Outlines

- Laplace Transform
- Exercise
- Laplace Transform Pairs for Common Signals:
- Region of Convergence
- Pole and zeroes
- Laplace Transform Application
- Exercise
- Properties of the Laplace Transform
- Exercise

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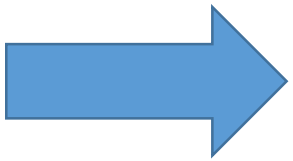
Laplace Transform (1/2)

- The Laplace Transform is a powerful tool for the analysis and design of CT LTI Systems
  - Characterize systems using the "Transfer Function"
  - Solve differential equations with non-zero initial conditions

- Unfortunately, the Fourier transform doesn't "converge" for some signals.

Fourier transform

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt} dt$$



Laplace transform

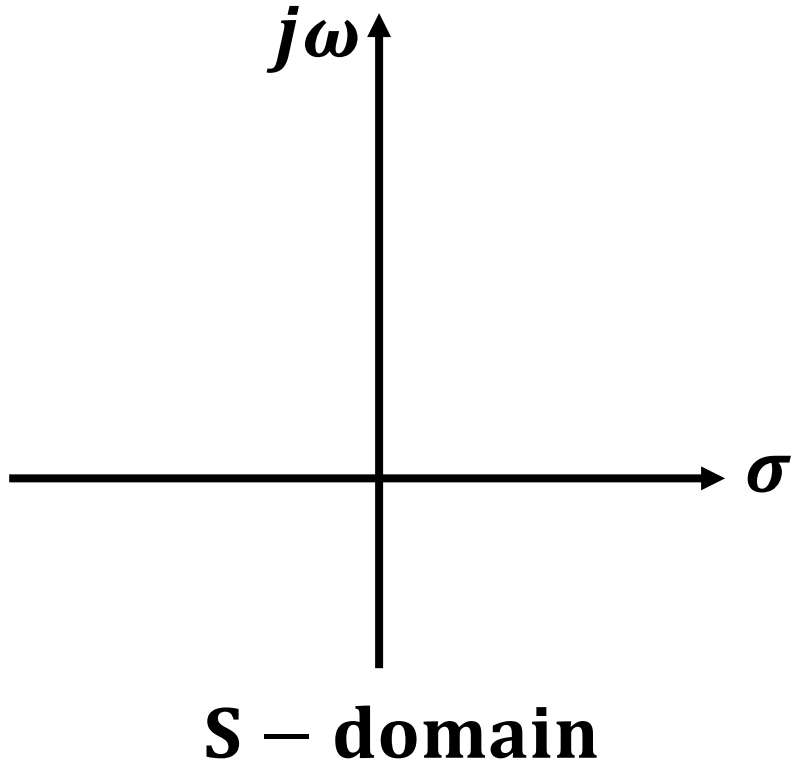
$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st} dt$$

Where, $s = \sigma + j\omega$

Here, the Laplace transform is introduced to represent continuous-time signals in the s-domain (s is a complex variable)

Laplace Transform (2/2)

- **s** is just a complex variable that we almost always view in **rectangular** form.



- For a general continuous-time signal *x*(*t*), the Laplace transform *X*(*s*) is defined as next:-

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

Called the *bilateral* (or *two-sided*) Laplace transform

$$X(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$$

unilateral (or *one-sided*) Laplace transform

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Exercise

Q1) Find the Laplace transform for $\mathbf{x(t) = \delta(t)}$

Ans.

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st} dt$$

$$X(s) = \int_{-\infty}^{\infty} \delta(t)\ e^{-st} dt$$

$$X(s) = \int_{-\infty}^{\infty} \delta(t)\ e^{-st} dt$$

$$X(s) = 1$$

$$X(t) = \delta(t) \Longleftrightarrow X(s) = 1$$

Q2) Find the Laplace transform for $\mathbf{x(t) = u(t)}$

Ans.

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st} dt$$

$$X(s) = \int_{-\infty}^{\infty} u(t)\ e^{-st} dt$$

$$X(s) = \int_{0}^{\infty} e^{-st} dt$$

$$X(s) = \frac{-1}{s}\, e^{-st}\Big|_0^{\infty} = \frac{-1}{s}[\, e^{-s(\infty)} - 1]$$

S > 0 to have a convergence function

$$X(s) = \frac{-1}{s}\, e^{-st}\Big|_0^{\infty} = \frac{1}{s}$$

$$u(t) \Longleftrightarrow \frac{1}{s}$$

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Laplace Transform Pairs for Common Signals:

- The Laplace transforms of some common signals are tabulated in Table 1.
- Instead of having to reevaluate the transform of a given signal, we can simply refer to such a table and read out the desired transform.

TABLE 3-1 Some Laplace Transforms Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	All s
$u(t)$	$\frac{1}{s}$	$\mathrm{Re(s)} > 0$
$-u(-t)$	$\frac{1}{s}$	$\mathrm{Re(s)} < 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$t^k u(t)$	$\frac{\mathrm{k!}}{s^{k+1}}$	$\mathrm{Re}(s) > 0$
$e^{-at} u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$
$te^{-at} u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$
$-te^{-at}u(-t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) < -\mathrm{Re}(a)$
$\cos \omega_0 tu(t)$	$\frac{s}{s^2+\omega_0^2}$	$\mathrm{Re}(s) > 0$
$\sin \omega_0 tu(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\mathrm{Re}(s) > 0$
$e^{-at} \cos \omega_0 tu(t)$	$\frac{s+a}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$
$e^{-at} \sin \omega_0 tu(t)$	$\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$

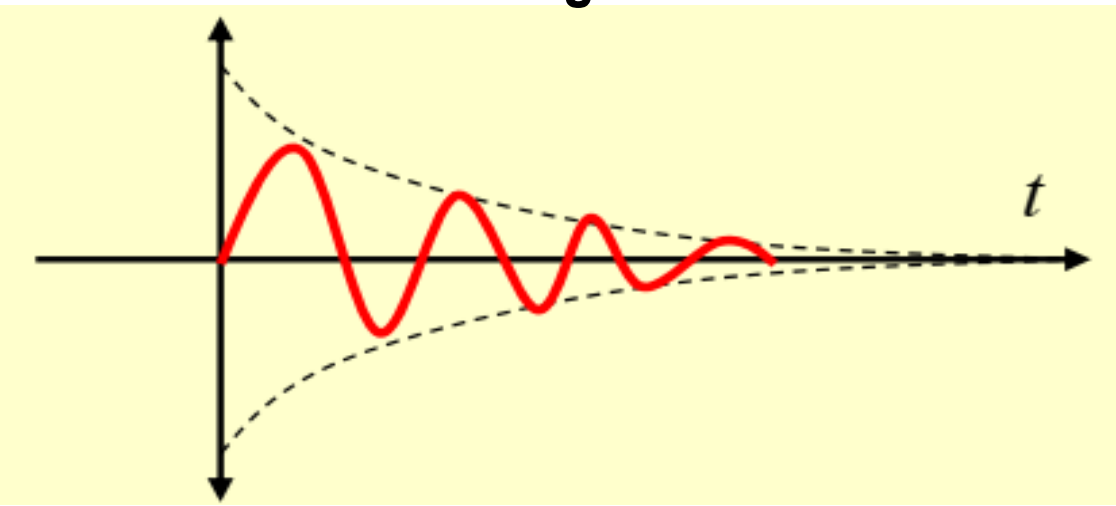
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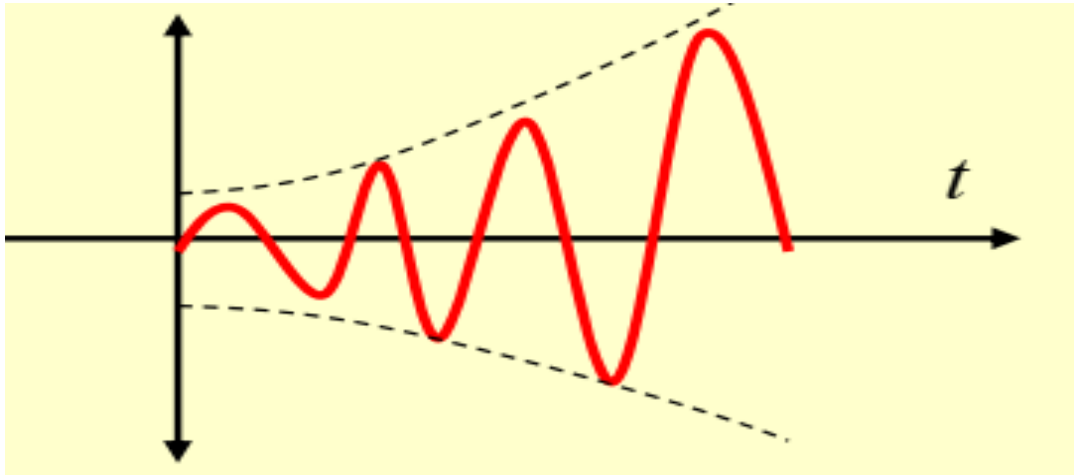
Region of Convergence (1/2)

- The range of values of the complex variables *s* for which the Laplace transform converges is called the *region of convergence* (ROC)
- Each signal has its own region of convergence.

Convergence



Un-Convergence



Region of Convergence (2/2)

Find the Laplace transform for $X(t) = e^{-at}\,u(t)$ where a>0

Ans.

The Laplace of x(t) is :-

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st} dt$$

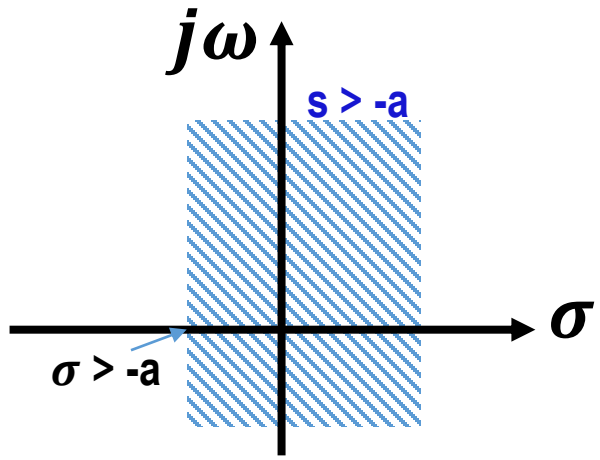
$$X(s) = \int_{-\infty}^{\infty} e^{-at}\, u(t)\, e^{-st} dt$$

$$X(s) = \int_{0}^{\infty} e^{-at}\, e^{-st} dt$$

$$X(s) = \int_{0}^{\infty} e^{-(s+a)t} dt$$

$$X(s) = \frac{-1}{s+a}\, e^{-(s+a)t}\Big|_{0}^{\infty}$$

$$X(s) = \frac{-1}{s+a}\left[\lim_{t\to\infty} e^{-(s+a)t} - 1\right]$$



s+a > 0 → s > -a to provide a convergence

$$X(s) = \frac{1}{s+a}$$

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Pole and zeroes (1/2)

- Usually, $X(s)$ will be a rational function in s; that is

$$X(s) = \frac{a_0 s^m + a_1 s^{m-1} + \cdots + a_m}{b_0 s^n + b_1 s^{n-1} + \cdots + b_n} = \frac{a_0}{b_0} \frac{(s - z_1) \cdots (s - z_m)}{(s - p_1) \cdots (s - p_n)}$$

- The coefficients a_0 to a_m and b_0 to b_n are are real constants.
- m and n are positive integers.

- The $X(s)$ is called a *proper* rational function if $n > m$, and an *improper* rational function if $n \leq m$.
- The roots of the numerator polynomial, Z_K, are called the *zeros* of $X(s)$ because $X(s)= 0$ for those values of s.
- Similarly, the roots of the denominator polynomial, p_k, are called the *poles* of $X(s)$ because $X(s)$ is infinite for those values of s.

Therefore, the poles of $X(s)$ lie outside the ROC since $X(s)$ does not converge at the poles, by definition

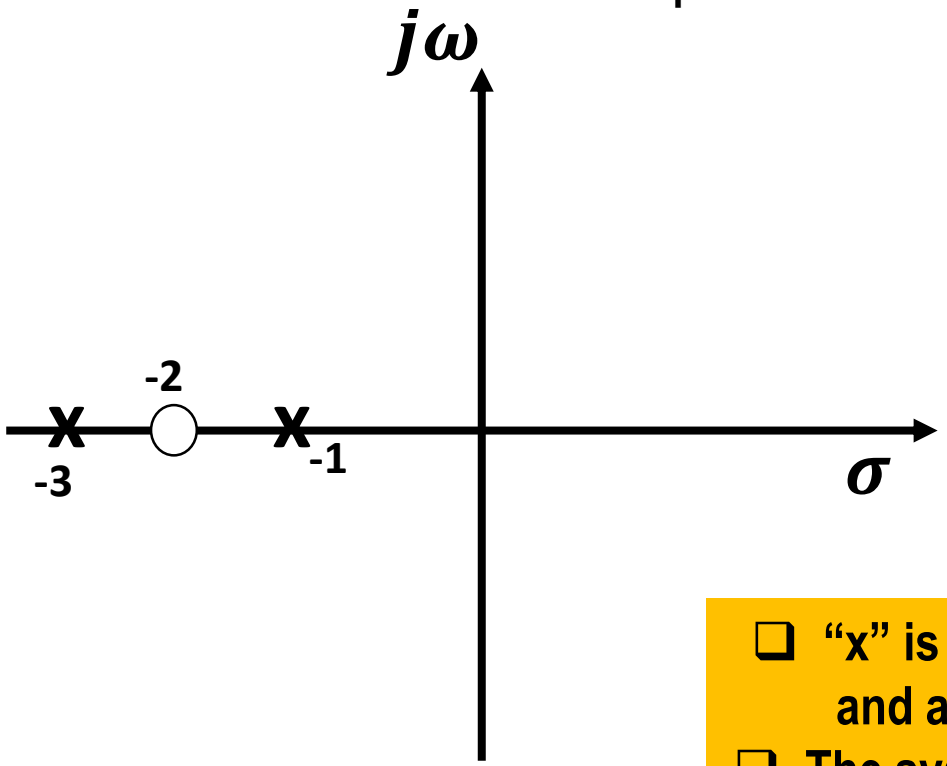
The zeros, on the other hand, may lie inside or outside the ROC

Pole and zeroes (2/2)

Ex., if $X(s) = \frac{s+2}{(s+1)(s+3)}$

One zero at s=-2

Two poles at s=-1 and s=-3

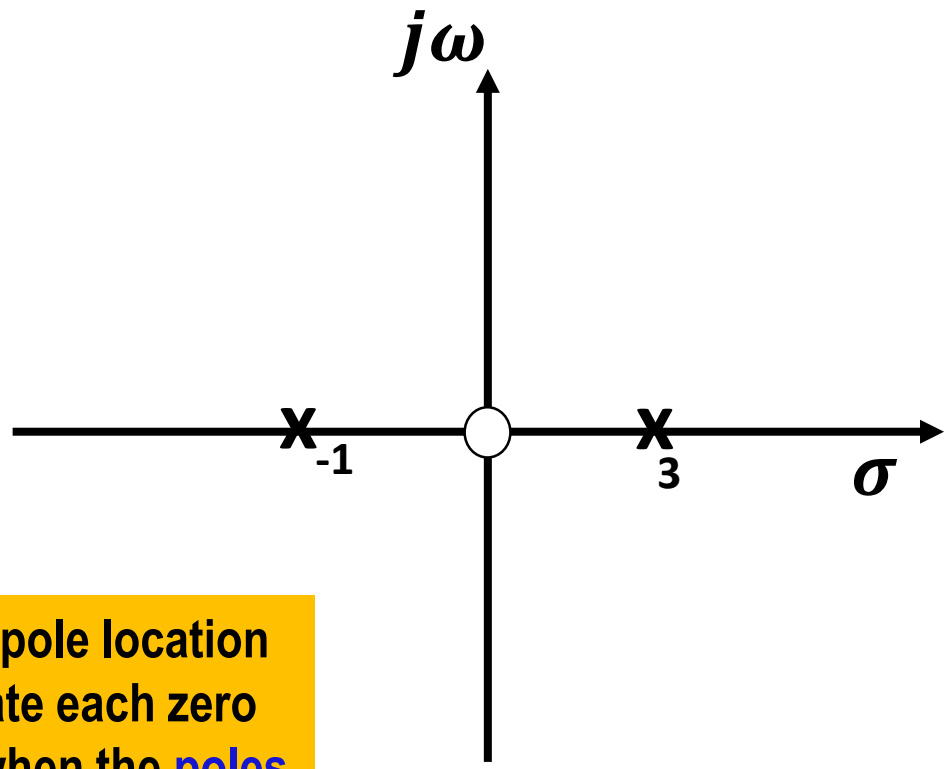


Stable

Ex., if $X(s) = \frac{s}{(s+1)(s-3)}$

One zero at s=0

Two poles at s=-1 and s=+3



Un-Stable

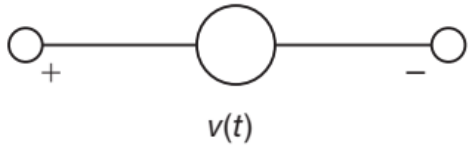
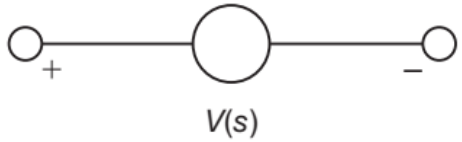
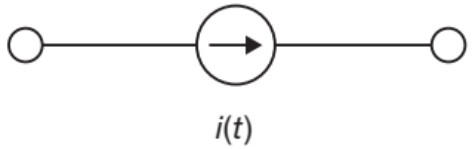
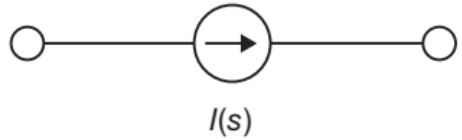
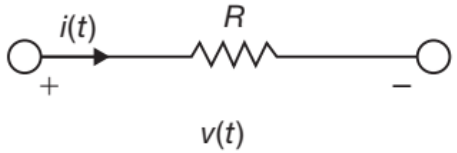
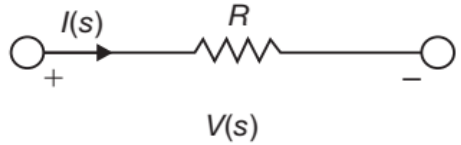
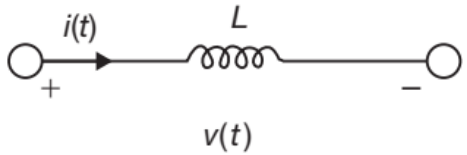
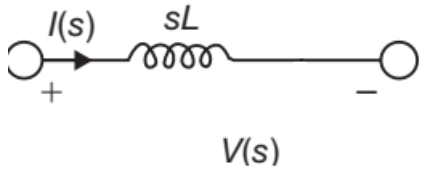
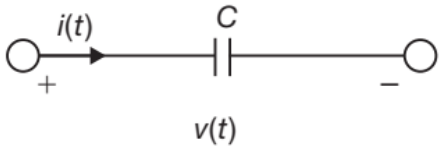
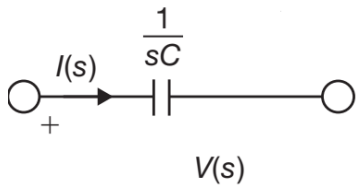
- "x" is used to indicate each pole location and an "o" is used to indicate each zero
- The system is called stable when the poles lies on the left side of s-plan

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Laplace Transform Application

- Laplace transform equivalents can be used to analysis the electric circuits

	Time domain	S- domain
Voltage source		
Current source		
Resistance		
Inductance		
Capacitance		

Laplace transform circuit-element models

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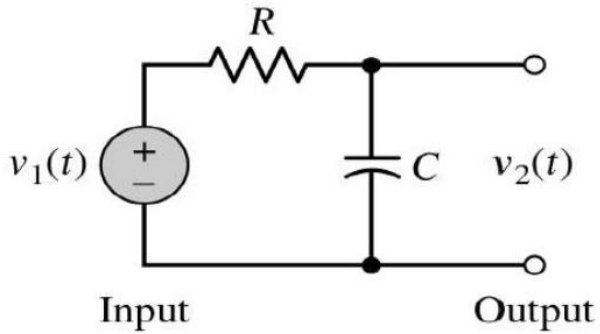
Exercise (1/3)

- Q1) For the low-pass-filter circuit,

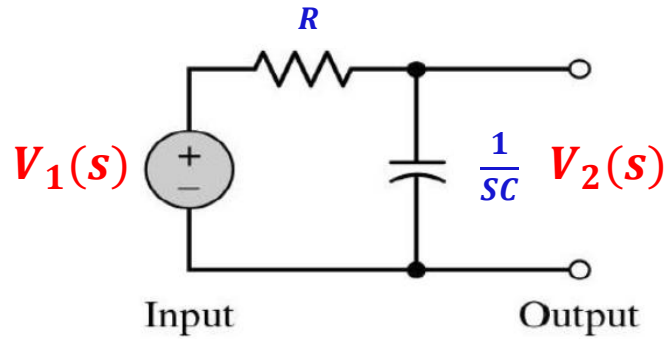
a) Derive the transfer function H(s)=$\frac{V_2(s)}{V_1(s)}$

b) Plot the s-domain with the poles and zeroes of the H(s)

c) Is the circuit stable or un-stable?



Ans.

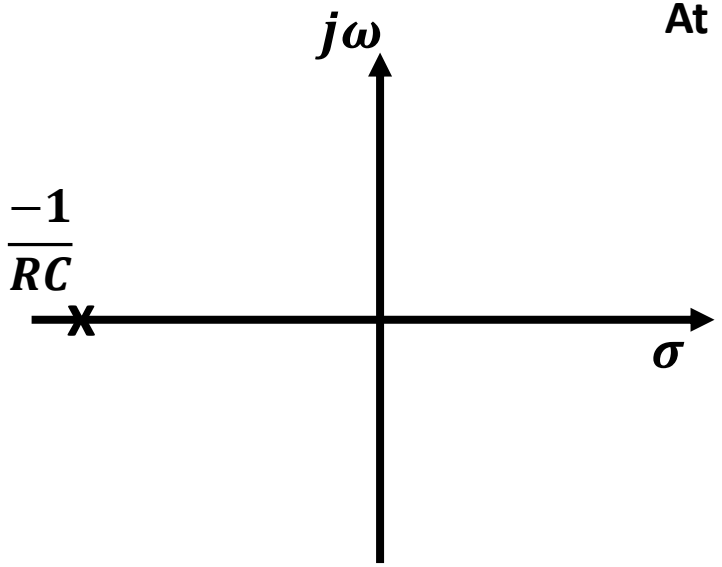


$$H(s)=\frac{V_2(s)}{V_1(s)}=\frac{\frac{1}{sC}}{\frac{1}{sC}+R}=\frac{1}{1+RSC}$$

No zeroes

There is a single pole

At p1= $\frac{-1}{RC}$



System is stable

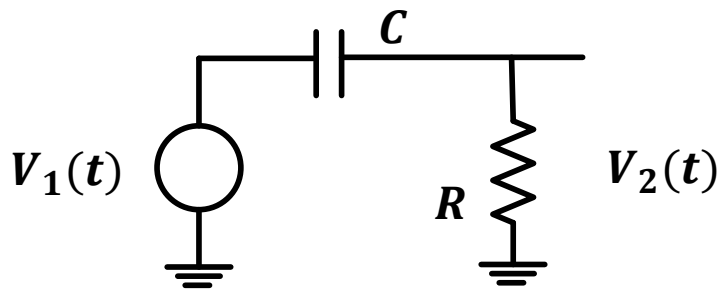
Exercise (2/3)

- Q2) For the high-pass-filter circuit,

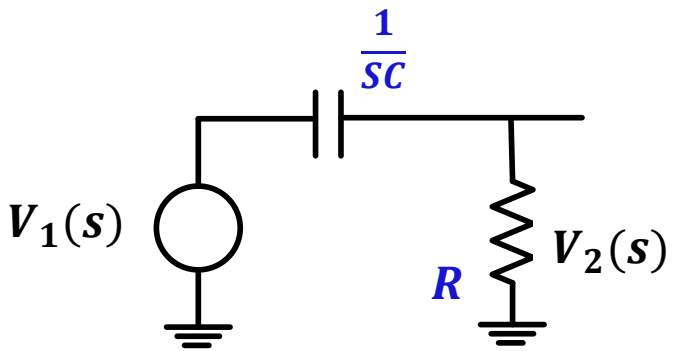
a) Derive the transfer function $H(s)=\frac{V_2(s)}{V_1(s)}$

b) Plot the s-domain with the poles and zeroes of the H(s)

c) Is the circuit stable or un-stable?



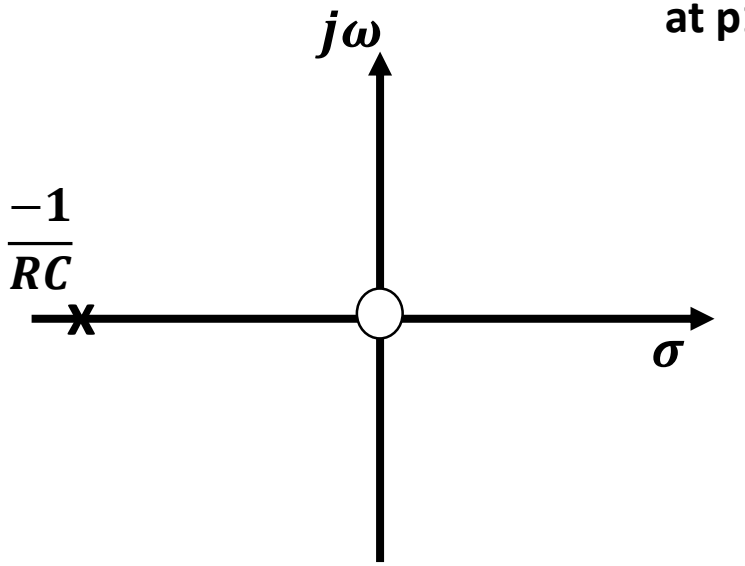
Ans.



$$H(s)=\frac{V_2(s)}{V_1(s)}=\frac{R}{\frac{1}{SC}+R}=\frac{SC}{1+RSC}$$

There is a single zero at Z1=0

There is a single pole at p1= $\frac{-1}{RC}$



System is Stable

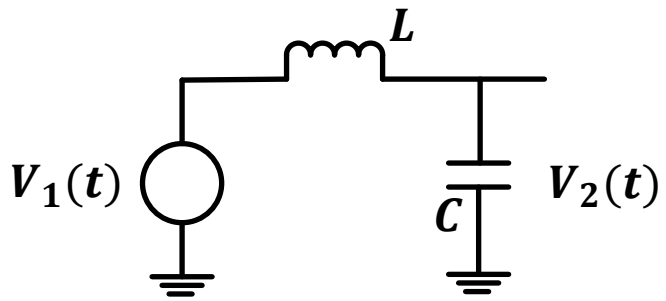
Exercise (3/3)

- Q3) For the oscillating circuit,

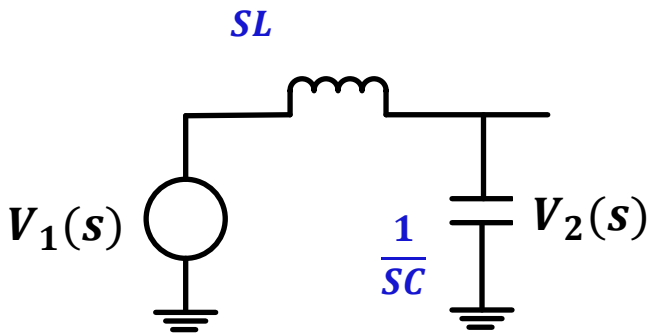
a) Derive the transfer function H(s)=$\frac{V_2(s)}{V_1(s)}$

b) Plot the s-domain with the poles and zeroes of the H(s)

c) Is the circuit stable or un-stable?



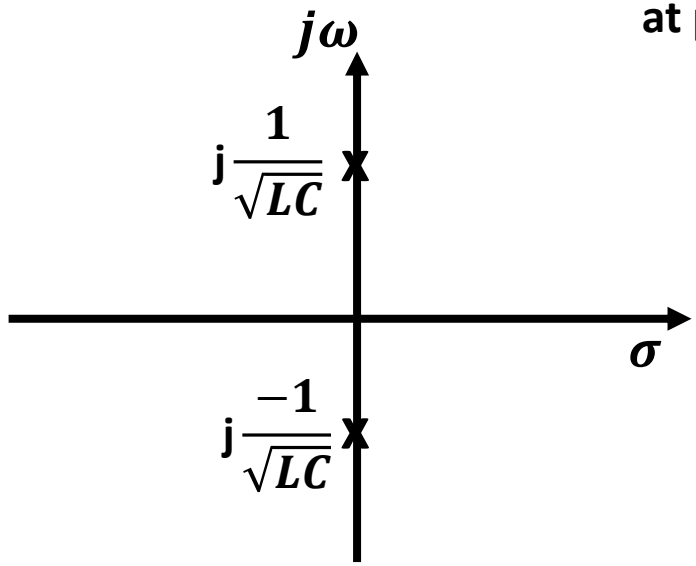
Ans.



$$H(s)=\frac{V_2(s)}{V_1(s)}=\frac{\frac{1}{SC}}{\frac{1}{SC}+SL}=\frac{1}{1+S^2LC}$$

No zeroes

There is double poles at p1= j$\frac{1}{\sqrt{LC}}$, p2= j$\frac{-1}{\sqrt{LC}}$



System is Critical

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Properties of the Laplace Transform

PROPERTY	SIGNAL	TRANSFORM
	$x(t)$	$X(s)$
	$x_1(t)$	$X_1(s)$
	$x_2(t)$	$X_2(s)$
Linearity	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(s) + a_2X_2(s)$
Time shifting	$x(t-t_0)$	$e^{-st_0}X(s)$
Shifting in s	$e^{s_0t}x(t)$	$X(s-s_0)$
Time scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(a)$
Time reversal	$x(-t)$	$X(-s)$
Differentiation in t	$\frac{dx(t)}{dt}$	$sX(s)$
Differentiation in s	$-tx(t)$	$\frac{dX(s)}{ds}$
Integration	$\int_{-\infty}^{t} x(\tau)\,d\tau$	$\frac{1}{s}X(s)$
Convolution	$x_1(t) * x_2(t)$	$X_1(s)\,X_2(s)$

A. Linearity

$$x_1(t) \leftrightarrow X_1(s)$$
$$x_2(t) \leftrightarrow X_2(s)$$
$$a_1x_1(t) + a_2x_2(t) \leftrightarrow a_1X_1(s) + a_2X_2(s)$$

B. Time Shifting

$$x(t) \leftrightarrow X(s) \qquad x(t-t_0) \leftrightarrow e^{-st_0}X(s)$$

C. Shifting in the s-Domain

$$x(t) \leftrightarrow X(s) \qquad e^{s_0t}x(t) \leftrightarrow X(s-s_0)$$

D. Time Scaling

$$x(t) \leftrightarrow X(s) \qquad x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{s}{a}\right)$$

E. Time Reversal

$$x(t) \leftrightarrow X(s) \qquad x(-t) \leftrightarrow X(-s)$$

F. Differentiation in the Time Domain:

$$x(t) \leftrightarrow X(s) \qquad \frac{dx(t)}{dt} \leftrightarrow sX(s)$$

G. Differentiation in the s-Domain

$$x(t) \leftrightarrow X(s) \qquad -tx(t) \leftrightarrow \frac{dX(s)}{ds}$$

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Exercise (1/3)

- Q1) if y(t)=x1(t)+x2(t) where x1(t)=$\delta(t)$ and x2(t)=u(t), find y(s) using the following table

TABLE 3-1 Some Laplace Transforms Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	All s
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$-u(-t)$	$\frac{1}{s}$	$\mathrm{Re}(s) < 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$t^k u(t)$	$\frac{k!}{s^{k+1}}$	$\mathrm{Re}(s) > 0$
$e^{-at} u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$

Ans.

Using linearity property

X1(t)= $\delta(t)$ → x1(s)=1

X2(t)= $u(t)$ → x2(s)=1/s

y(t)= x1(t)+x2(t) → y(s)=1+(1/s) =$\frac{s+1}{s}$

Note:
If the table is not given, you must het x1(s) and x2(s) by Your self as in the beginning of the lec.

Exercise (2/3)

- Q2) if y(t)=x(t-to) where x(t)= t u(t), find y(s) using the following table

TABLE 3-1 Some Laplace Transforms Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	All s
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$-u(-t)$	$\frac{1}{s}$	$\mathrm{Re}(s) < 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$t^k u(t)$	$\frac{k!}{s^{k+1}}$	$\mathrm{Re}(s) > 0$
$e^{-at} u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$

Ans.

Using time shifting property

X(t)= t u(t) → x(s)= $\frac{1}{s^2}$

X(t-to)= (t-to) $u(t-to)$ → x(s)=e^{-sto} $\frac{1}{s^2}$

y(t)= X(t-to) → y(s)= e^{-sto} $\frac{1}{s^2}$

Note:
If the table is not given, you must het x1(s) and x2(s) by Your self as in the beginning of the lec.

Exercise (3/3)

- Q3) if y(t)=x1(t)*x2(t) where x1(t)=t u(t) and x2(t)=u(t), find y(s) using the following table

TABLE 3-1 Some Laplace Transforms Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	All s
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$-u(-t)$	$\frac{1}{s}$	$\mathrm{Re}(s) < 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$t^k u(t)$	$\frac{k!}{s^{k+1}}$	$\mathrm{Re}(s) > 0$
$e^{-at} u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -\mathrm{Re}(a)$

Ans.

Using convolution property

X1(t)= t u(t) → x1(s)= $\frac{1}{s^2}$

X2(t)= $u(t)$ → x2(s)= $\frac{1}{s}$

y(t)= x1(t)*x2(t) → y(s)= x1(s) x2(s)= $\frac{1}{s^2}\frac{1}{s}=\frac{1}{s^3}$

Note:
If the table is not given, you must het x1(s) and x2(s) by Your self as in the beginning of the lec.

Thanks!